

Summary

Software Engineer with an M.S. in Electrical Engineering, experienced in frontend/backend platforms, AI services, and robotics software from requirements and architecture through Docker deployment, testing, and real-world integration. Skilled in Python, C++, C#/.NET, React/TypeScript, FastAPI, WebSocket, ROS 2, Isaac Sim/Lab, Isaac GROOT, computer vision, and Digital Twin workflows.

Focus Areas

70% → 90% cable-insertion success	3-4x manual data volume in the same time window	-2 operators manual staffing needs
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Experience

AMTRAN TECHNOLOGY CO., LTD. | AI Robotics Engineer May 2025 - Present | New Taipei, Taiwan

- Owned software delivery from requirements and system architecture through frontend/backend implementation, Docker deployment, testing, on-site integration, and maintenance for AI and robotics systems.
- Built a Next.js / React Flow / FastAPI / WebSocket workflow and data-recording platform for automated task execution and AI / VLA data workflows, reaching approximately 3-4x the manual data volume in the same time window and reducing manual staffing needs by two operators.
- Built a React / TypeScript / FastAPI / ROS 2 web control and monitoring platform with real-time feedback, video, system status, control commands, and Docker / Nginx deployment; improved cable-insertion success from 70% to 90%.
- Integrated ROS 2 control and simulation, Isaac GROOT N1.7 data conversion, model training, policy inference, visual perception, and PatchCore anomaly detection across simulation and physical-robot workflows.

ISCOM | Software Engineer Intern Feb 2022 - Jun 2022 | Kaohsiung, Taiwan

- Participated in backend API development, frontend integration, and web feature implementation using C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git.

Skills

Software Engineering & Platforms	Python, C++, C# / .NET, FastAPI, React / TypeScript, Next.js, WebSocket, REST APIs, Docker, Linux
AI / ML & Computer Vision	PatchCore anomaly detection, PyTorch, OpenCV, YOLO OBB, AprilTag, RealSense, PyQt
Robotics & Real-Time Systems	ROS 2, NVIDIA Isaac Sim / Lab, MoveIt 2, cuMotion, Isaac GROOT N1.7, VLA, Policy Inference, Digital Twin
Systems Integration & IoT	frontend/backend platforms, simulation / physical validation, Arduino, ESP32, MQTT

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
- B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
- Honors: RHCSA; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place